



2025

U22

BOOKLET NO.

158180

Mechanical Engineering
Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use scientific (non-programmable type) calculators.

P.T.O.

**SECTION – A**

Q1. Answer the following questions in about **150 words each**. **(5×10=50)**

Q1. (a) A 150 mm diameter valve against which a steam pressure of 1.5 MN/m^2 is acting is closed by means of a square threaded screw. The external diameter of the screw is 50 mm with 8 mm pitch. If the coefficient of friction is 0.15, find the torque required to turn the handle. Assume uniform stress distribution and neglect self weight of member. **10**

Q1. (b) A mild steel rod is designed to carry an axial tensile load of 75 kN. The permissible tensile stress for the material is 100 MPa. Determine the required cross-sectional dimensions of the member for each of the following cases. **10**

- i) When the member is of circular cross-section, find the diameter.
- ii) When the member is of square cross-section, find the side of the square.
- iii) When the member is of rectangular cross-section, where the width is twice the thickness, determine both the width and thickness.

Q1. (c) Explain the structure and electronic properties of Carbon Nano Tubes (CNTs). How does chiral vector influence whether a nanotube is metallic or semiconducting ? **10**

Q1. (d) i) Distinguish between the cycloidal and Involute gear profiles along with tooth forming methods. **5**

ii) Explain the difference between Flywheel and Governor with suitable examples. **5**

Q1. (e) Explain the transformation of austenite to martensite in steels. Discuss the main characteristics and morphology formed during this transformation. **10**

Q2. (a) i) A cylindrical pressure vessel has an internal diameter of 600 mm and a wall thickness of 25 mm. It is subjected to an internal fluid pressure of 2.5 N/mm^2 . Determine :

- i) The hoop (circumferential) stress
- ii) The longitudinal (axial) stress and
- iii) The maximum shear stress developed in the cylinder wall



Assume that the vessel satisfies the condition of a thin cylinder (i.e. thickness $< 1/20$ of the diameter) and that the internal pressure is uniformly distributed.

10

- ii) A solid circular steel shaft is transmitting power in a mechanical system. The shaft experiences a bending moment of 3500 Nm and torsional moment of 12000 Nm. The material of the shaft is medium carbon steel with an ultimate tensile strength of 720 MPa and ultimate shear strength of 520 MPa. If the design requires factor of safety of 5, determine the minimum diameter of the shaft to ensure safe operation.

10

- Q2. (b) i) Define the Atomic Packing Factor (APF).

Derive the expression for the volume of the unit cell and the atomic packing factor for a Body-Centred Cubic (BCC) crystal structure in terms of the atomic radius R .

10

- ii) Explain the microstructural changes that occur during the slow cooling of hypoeutectoid steels. Discuss the formation of different phases and their effect on mechanical properties.

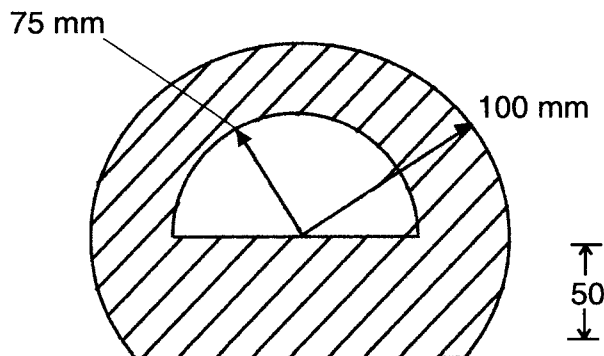
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- Q2. (c) A car of mass 1200 Kg moves along a curve having curvature of 50 m. The speed of the car increases uniformly from 12 m/s to 20 m/s over a distance of 100 m. Determine tangential acceleration of the car. Also determine angular velocity and centripetal acceleration of the car at the final speed. Assume that the road is flat and the coefficient of friction between the surface of tires and road is 0.5. What will be the maximum frictional force between the tires and the road ?

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- Q3. (a) i) A semicircular section is removed from the plane area as shown in fig. Find the centroid of the remaining shaded area.

12



- ii) Explain the concept of equation of equilibrium with suitable example.

8

- Q3. (b) i) What is the whirling speed of a shaft ? Explain by deriving the suitable equation.

10

- ii) Four masses A, B, C and D are carried by a rotating shaft at radius 150, 125, 200 and 150 mm respectively. The planes in which the masses revolve are spaced 500 mm apart. The masses of B, C and D are 10 kg, 12 kg and 8 kg respectively. Find the required mass A and angular position of four masses so that the shaft shall be in complete balance.

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- Q3. (c)** A cast iron hollow cylinder has an outer diameter of 95 mm and inner diameter of 75 mm. The cylinder is subjected to an axial compressive load of 75 kN. For a length of 700 mm calculate :
- i) The total axial shortening of the cylinder.
 - ii) The normal (axial) stress produced by the load.
- Use $E = 100 \text{ GPa}$ and neglect any buckling or instability effects. **10**
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- Q4. (a)** i) Differentiate between normal stress and shear stress. **7**
- ii) A rectangular beam 100 mm wide and 150 mm deep is subjected to a shear force of 30 kN. Determine
- i) Average shear stress.
 - ii) Maximum shear stress. **6**
- iii) Explain the principle of plane motion for a rigid body and describe how velocity and acceleration of a point are determined. **7**
- Q4. (b)** i) Explain the purpose and processes of full annealing in steels. How are the derived mechanical properties achieved through this process ? **10**
- ii) What is the purpose and stages of normalizing in steels ? What are the desired mechanical properties to be achieved through normalizing and how are they achieved ? **10**
- Q4. (c)** Compare the stress-strain behavior of mild steel and copper. Explain the need for determining proof stress and yield stress in engineering materials. **10**
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SECTION – B

- Q5.** Answer the following questions in about **150** words **each**. **(5×10=50)**
- Q5. (a)** Following data operates for an Ultrasonic Machining (USM) process.
- Amplitude of vibration : 30 μm
Frequency of vibration : 20 kHz
Diameter of abrasive grain : 0.015 mm
Static feed force : 5 N
Volumetric concentration of abrasive particles in a slurry : 50%
Ratio of tool indentation to work indentation : 5
Hardness of the workpiece : 7000 N/m^2
Determine the time required to machine a square block of side 6 mm and thickness 5 mm. **10**
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- Q5. (b)** For a particular tool-work combination, it is observed that, for a cutting speed of 200 m/min the tool life was 30 minutes while for a cutting speed of 300 m/min the tool life was 12 minutes. Find the tool life for cutting speed of 250 m/min and also determine the cutting speed that will give tool life of 25 minutes. **10**
- Q5. (c)** The selling price of two products P_1 and P_2 is Rs. 250/- and Rs. 360/- each while the variable prices are 80 and 150. Sales Mix ($P_1 : P_2$) is 4 : 3 and the expected sale is 7000 units. If the fixed cost is Rs. 3,20,000. Determine :
- i) Break even Point
 - ii) Percent Margin of Safety.
 - iii) Effect on break even point if selling price of product P_1 is raised by 15%. **10**
- Q5. (d)** Explain the concept of production scheduling and discuss any two production scheduling algorithms. **10**
- Q5. (e)** The length of a work piece is measured using a micrometer is 50.12 mm. The temperature during the measurement was 35°C. Calculate the true length of the component. Also determine the percentage error if the zero error is +0.02 mm, the standard temperature is 20°C, and coefficient of thermal expansion is $20 \times 10^{-6}/^\circ\text{C}$. **10**
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- Q6. (a) i)** Explain Merchant force circle along with its assumptions. **5**
- ii) In an orthogonal cutting, the power required for cutting is 1900 W, cutting speed is 120 m/min, measured thrust force is 350 N. Calculate the shear angle to limit the shear force to 1000 N and rake angle to limit the friction force to 700 N. **15**
- Q6. (b) i)** Explain the concept of lean management and discuss following techniques with suitable examples :
- i) 5S
 - ii) Kaizen
 - iii) Poka-Yoke
 - iv) Kanban. **10**
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- ii) An operation having five work elements is analysed using Pre determined Motion Time Study (PMTS). The analyst assigned following TMU (Time Measurement Units) values from PMTS Table :

Sr. No.	Motion Element	TMU
1	Reach for part	20
2	Grasp part	12
3	Move to fixture	30
4	Position part	10
5	Release part	08

Consider the allowances as follows :

Personal allowance = 5%

Contingency allowance = 2%

Fatigue allowance = 3% (applicable to motion element Sr. No. 3 only)

Calculate how many complete parts can be produced in a hour assuming that operation time is negligible.

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- Q6.** (c) Determine the maximum possible welding speed for a butt arc welding process having arc power 4.2 KVA and is used to weld two steel plates each of 2.5 mm thickness

Given that :

Fraction of total time during which the arc is on = 85%

Thermal diffusivity of steel = $1.2 \times 10^{-5} \text{ m}^2/\text{sec}$

Thermal conductivity of steel = 43.6 W/m°C

Melting point of steel = 1500°C

Ambient temperature = 20°C

Minimum width of weld to be maintained is = 3.45 mm.

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- Q7.** (a) i) Explain in detail the working principle, construction and applications of the following gauges.

A) Plug gauge

B) Snap gauge.

10

- ii) Explain five important differences between jigs and fixtures. Also describe how jigs and fixtures improve productivity in mass production.

10



- Q7. (b) i) Explain the probabilistic inventory model when demand and lead time are variables. 5

- ii) Quarterly demand for a product is as shown in the table below. Calculate the safety stock and reorder point for each quarter if :

Ordering cost per order = Rs. 200

Holding cost = Rs. 5 per unit per year

Lead time = 10 days

Target service level = 95% ($z = 1.645$)

Coefficient of variation for each quarter (CV) = 0.25.

Quarter	Demand (Units)	Days in Quarter	
Q1	3000	90	
Q2	2500	91	
Q3	4000	92	
Q4	2500	92	15

- Q7. (c) A manufacturing firm produces product with targeted dimension of 25 unit. The observed values for 6 samples with 5 measurements each are given below :

Sample 1 : {24, 23, 26, 21, 24}

Sample 2 : {22, 25, 23, 23, 27}

Sample 3 : {26, 24, 25, 23, 21}

Sample 4 : {25, 27, 26, 22, 23}

Sample 5 : {21, 22, 23, 24, 25}

Sample 6 : {24, 27, 26, 24, 25}

Determine out of control points on π chart and also compute C_p and C_{pk} .

For subgroup 5 (sample size), $A_2 = 0.577$ and $d_2 = 2.326$. Consider Lower Specification Limit (LSL) = 22 and Upper Specific Limit (USL) = 28.

Interpret your results. 10



- Q8. (a)** i) Explain the role of friction in strip rolling operation. 5
- ii) A strip with cross section of $150 \text{ mm} \times 6 \text{ mm}$ is being rolled with 20% reduction of area, using 400 mm diameter steel rolls. Before and after rolling, the shear yield stress of the material is 0.35 KN/mm^2 and 0.4 KN/mm^2 respectively. Assuming the coefficient of friction to be 0.1, determine the location of the neutral point. 15

- Q8. (b)** A department store is trying to determine the amount of lost sales while it was shut down in month of July and August due to floods. The past sales in the year are as follows :

Month	Sales (in thousand Rs.)
January	185.72
February	167.84
March	205.11
April	210.36
May	255.57
June	261.19

- i) Use Exponential smoothening method with $\alpha = 0.4$ to develop forecast for July and August .
- ii) Use trend projection to forecast sales for July and August.
- iii) The insurance company has proposed a settlement of Rs. 2,40,000 for sales lost in July and August. Is the amount fair ? If not, what amount is expected ? Assume January sales as forecast for period one. In exponential method, sale projected for July and August may be same. 20

- Q.8. (c)** Show that the average extrusion pressure (P_e) in extrusion process is :

$$P_e = \sigma_f \left[\ln R + \frac{4\mu L}{D} \right]$$

Where σ_f = Flow stress of material

R = Extrusion ratio

μ = Coefficient of friction between billet and container wall

L = Billet length and

D = Billet diameter. 10